ALMON

Exploring Labour Market Trends: Austria and Neighbouring Countries (2014 - 2024)

Final project report

Submitted by

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Content

Project Overview

The ALMON project has been developed as part of the IP: SDI course of the Applied Geoinformatics Master's programme in the winter term 2024/25. This first chapter provides an overview of the project. This includes general information, the project's practical context, the stakeholders, the target group, goals and objectives within the scope of the project.

1.1 General information

Project		
Acronym	ALMON	
Title	Exploring Labour Market Trends: Austria and Neighbouring Countries (2014–2024)	
Duration	Start: 03.03.2025	End: 30.06.2025
Authors	Ekata Leo Oni, Ce Li	
Contractor	Arbeitsmarktservice, Roxana Mînzatu (European Commissioner for Jobs and Social Rights)	

1.2 Context

This project builds a spatial data infrastructure (SDI) to analyze and compare labour market trends across Austria and its bordering countries, including Germany, Italy, Slovenia, Hungary, Slovakia, Czechia, and Switzerland. Using multi-year employment data from Eurostat, the project supports interactive comparisons across space and time, focusing on activity status, gender, and age groups.

1.3 Concept and Objectives

This project offers a reusable framework for analyzing regional employment trends across central Europe. It serves:

- Labour market policy analysts (e.g., AMS Austria, regional agencies)
- European commission
- Researchers and open data users seeking reliable cross-country datasets

Benefits include:

- Accessible dashboards and web services for cross-country comparison.
- Standardized and documented geospatial datasets.
- Awareness and data-driven insight into employment dynamics over time.

1.4 Non-goals

- No forecasting or predictive modelling
- No sub-national analysis (i.e., no municipal/provincial detail)
- No integration of economic indicators beyond labour force statistics

1.5 Stakeholders

ALMON project is a navigation and tracking system for Arbeitsmarktservice and for the people who work in the European Commissioner for Jobs and Social Rights office. Both of them will receive the results and use them for future prediction and decision-making. The aim is to achieve a better understanding of the labour market between Austria and its nearby countries.

1.6 Methodology

This project employs a spatial data infrastructure (SDI) framework to facilitate structured access to Austria's 2022 labour market statistics. The methodological workflow consists of six core phases: data acquisition, pre-processing, spatial organization, infrastructure design, dissemination, and visualization. Official datasets from Statistik Austria are cleaned and standardized using Python, followed by geocoding with verified administrative boundaries. The processed data are integrated into a PostGIS database, accompanied by ISO 19139-compliant metadata. Geospatial web services (WMS/WFS) are then configured using GeoServer to ensure OGC-compliant dissemination. A user-oriented dashboard is developed via ArcGIS Online to support interactive exploration of labour patterns by gender, age, and region. Throughout the process, the team applies open-source tools and adheres to FAIR data principles, ensuring transparency, reusability, and long-term interoperability. The methodology is designed to empower policy stakeholders such as AMS advisors by enabling dynamic, evidence-based analysis of Austria's employment landscape.

1.7 Up to date Status

The foundation for the project has been set. We've outlined our workflow, clarified our objectives, and agreed on how tasks will be shared within the team. The main dataset from Statistik Austria is already in hand, and we've started reviewing it to understand its structure and identify any gaps or issues. We're in the process of preparing our database environment and planning how to geocode and clean the data. With the core tools and timelines now in place, we're ready to move from planning into hands-on development over the coming weeks.

1.8 Project Setting

The project will be carried out remotely / in person, with regular coordination between team members through digital platforms. Development and testing will take place using open-source tools and institutional infrastructure. Key outputs such as web services and the dashboard will be hosted online to ensure accessibility for stakeholders throughout and beyond the project period.

2 Work Plan

The work plan is presented below (Figure 1). Five work packages constitute the project, from overall project management **(WP1)**, data discovery and preparation **(WP2, WP3)** to the final stages, including service creation and data sharing **(WP4, WP5), and Visualization (WP6).**

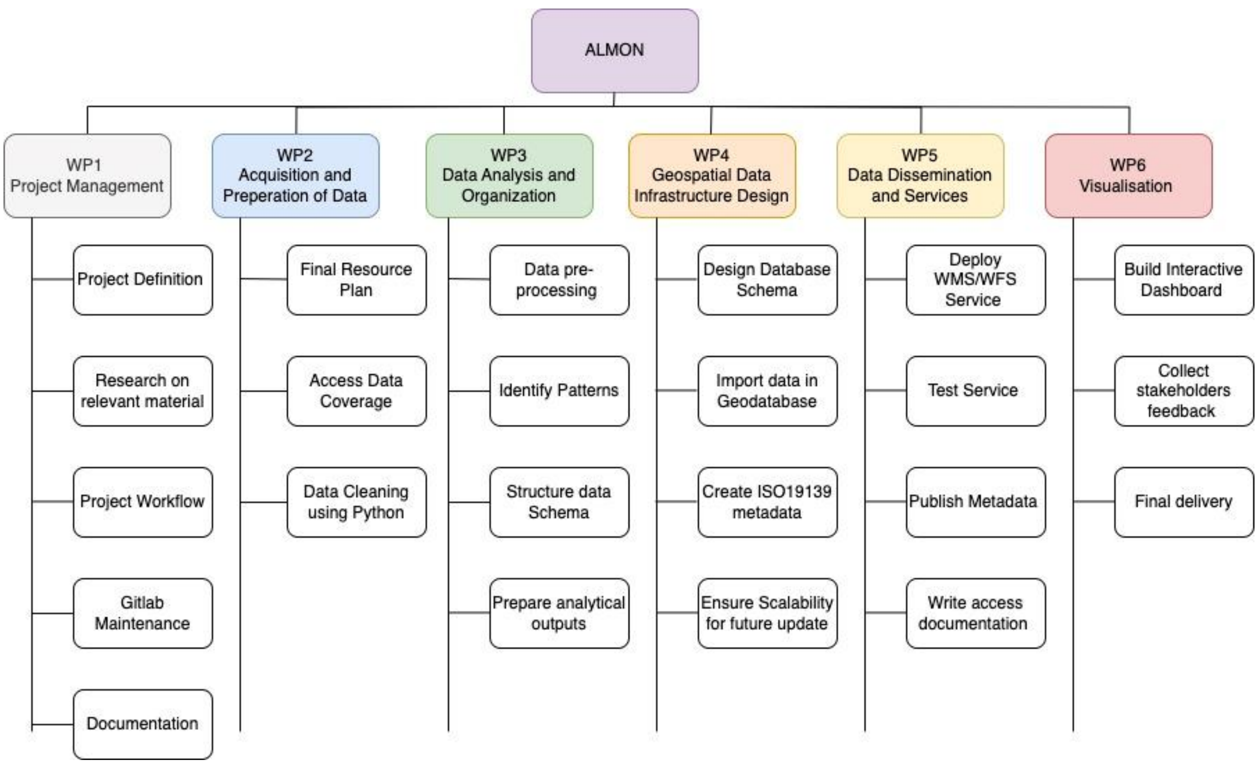


Figure 1: Work plan break down structure.

2.1 WP1: Project Management

Objectives and tasks

Establish a well-structured workflow and ensure smooth execution of all project phases. This includes planning, coordination, reporting, and data preparation steps such as data cleaning, uploading, and feature creation. The following objects and tasks:

- Define project goals, milestones, and responsibilities.
- Set up GitLab for collaboration and progress tracking.
- Establish a timeline and meeting schedule.
- Compile mid-term notes and submit final project report.

Gantt chart

The Gantt chart (Table 1) is a common visualization and scheduling tool for project management. It helped to map out all WPs and tasks of the ALMON project.

Tasks	Responsible	Start	End	Days	Status	3/3	3/6	3/9	3/12	3/20	3/24	3/30	4/5	4/10	4/15	4/25	5/5	5/10	5/15	6/1	6/2	6/15
Project Management						Start																
Gitlab Update	Ekata Leo Oni	3-Mar	6/30	119	in progress																	
Documentation	CE LI	3-Mar	6/30	119	in progress																	
Set kick-off meeting	Ekata Leo Oni	3-Mar	3/3	0	Complete																	
Agree on objectives	CE LI	3-Mar	6-Sep	187	Complete																	
Topic and Data						Start																
Choice of Topic	Oni, CE	3-Mar	6-Mar	3	Complete																	
Data research	Oni	6-Mar	24-Mar	18	Complete																	
Final Resource Plan	CE	9-Mar	24-Mar	15	Complete																	
Data Preperation										Start												
Data cleaning & analysis	Oni, CE	24-Mar	5-Apr	12	Complete																	
Data Pre-processing	CE	30-Mar	15-Apr	16	Complete																	
Infrustructure design															Start							
Integration into Database	Oni	15-Apr	5-May	20	Complete																	
Publish OGC Web Service	Oni, CE	5-May	10-May	5	Complete																	
Test and refine	Alex B.	5-May	1-Jun	27	Complete																	
Visualization																				Start		
Choice of Platform	Oni, CE	1-Jun	2-Jun	1	Complete																	
Dashboard Design	CE	1-Jun	30-Jun	29	Complete																	
Data Integration	Oni	1-Jun	30-Jun	29	Complete																	
Launch		30-Jun	30-Jun																			

Table 1: Gantt chart of the ALMON project

GitLab

The GitLab page of project ALMON features a wiki, for documentation, and issues that were regularly updated, this helped for keeping track of open tasks. The repository also contains shared documents, data, and metadata.

2.2 WP2: Acquisition and Preparing of Data

Objectives and tasks

Identify, collect, and prepare employment data for geospatial analysis.

Workflow

- Acquire labour market data from Statistik Austria (AEST 2022)
- Assess data coverage, format, and quality.
- Clean and standardize datasets using Python.
- Geocode data and verify coordinate accuracy.

Area of Interest

This project investigates labour market trends across Austria and its neighbouring countries (Germany, Italy, Slovenia, Hungary, Slovakia, Czechia, and Switzerland) over the 2004 - 2024 period. It focuses on spatial and demographic disparities in employment indicators, particularly activity status segmented by gender and age groups. The project aims to transform underutilized Eurostat employment data into actionable insights through the development of a standardized spatial data infrastructure (SDI), supporting policy design, educational initiatives, and cross-border research.

Data Acquisition

Labour market data were primarily sourced from **Eurostat,** covering the period from **2004 to 2024**, with a focus on employment rates and activity status disaggregated by **gender, age group,** and **country**. Supplementary data were obtained from **Statistik Austria (AEST 2022)** to ensure national completeness and temporal continuity.

All datasets underwent an initial quality assessment to evaluate coverage, consistency, and classification standards. Cleaning procedures included the removal of duplicates, harmonization of attribute formats, and handling of missing values. Subsequently, data were **geocoded** using national administrative boundaries and standardized using common geometries from **Eurostat** and **Natural Earth**. The cleaned datasets were structured for integration into a **PostGIS** spatial database, supporting further spatial analysis and visualization.

This systematic acquisition and preparation phase ensured the datasets were interoperable, reusable, and suitable for deployment within a geospatial data infrastructure (SDI) aligned with **ISO 19139** metadata standards

2.3 WP3: Data Analysis and Organization

Objectives and tasks

Organise, analyse, and structure data for SDI integration.

- Check data quality and identify possible challenges for its processing and overall usage.
- Decide on software to be used for data preparation (e.g., Python or R)
- Prepare/clean the data
- Import data into PostGIS database
- Prepare metadata

Workflow

- **Data Sourcing**
  - Acquire employment-related datasets from Eurostat and Statistik Austria (AEST 2022)
  - Select indicators: employment rate, labour force participation, segmented by age, gender, and country
  - Confirm temporal coverage (2004 - 2024) and availability across all target countries
- **Initial Quality Assessment**
  - Examine data completeness, granularity, and reporting intervals
  - Identify missing values, duplicate records, inconsistent units, or mismatched categories
- **Data Cleaning** *(Python-based processing)*
  - Standardize variable names and data types
  - Handle missing values through interpolation or annotation
  - Normalize indicator definitions to resolve cross-national classification discrepancies
- **Geocoding**
  - Assign spatial identifiers using national administrative units
  - Apply standardized geometries from Eurostat and Natural Earth
  - Validate geographic consistency and resolve topology mismatches
- **Attribute Harmonization**
  - Align classification schemas (e.g., age brackets, employment status) across countries
  - Create a unified schema for gender, age group, and temporal breakdown

- **Dataset Structuring**
  - Reorganize the cleaned data into relational tables suitable for PostGIS integration
  - Prepare schema definitions for seamless linking with spatial geometries
- **Documentation**
  - Record preprocessing steps, variable mappings, and data transformations
  - Generate interim metadata descriptions for later ISO 19139 compliance

Integration into database

Following data cleaning and geocoding, the standardized employment datasets were integrated into a PostGIS spatial database. A relational schema was designed to accommodate temporal, demographic, and spatial dimensions, enabling efficient querying and analysis. Spatial joins were performed to link attribute data with national boundary geometries, ensuring full geospatial alignment. Metadata structures were created in accordance with ISO 19139 standards, facilitating discoverability and interoperability. The resulting database forms the core of the project's spatial data infrastructure (SDI), supporting downstream services and visualization layers.

2.4 WP4: Geospatial Data Infrastructure Design

Objectives and tasks

Build a PostGIS database and define SDI architecture.

Workflow

The SDI design phase focuses on constructing a robust geospatial backend to support data dissemination and visualization. This begins with designing a PostGIS database schema tailored to accommodate multi-country employment data with temporal and demographic granularity. Cleaned and geocoded datasets are imported into the database, with spatial indexing applied to optimize performance. Parallelly, ISO 19139-compliant metadata are generated to document data provenance, structure, and usage constraints. The architecture is built with scalability in mind, allowing for future updates or indicator expansion. The finalized SDI is then connected to a GeoServer instance to enable standards-based dissemination through WMS/WFS services, laying the groundwork for external access and web-based visualization.

2.5 WP5: Data Dissemination and Service

Objectives and tasks

Develop and publish open geospatial services for external access.

Workflow

This phase operationalizes access to the processed labour market data through standardized geospatial web services. It begins with the configuration of GeoServer**,** which is connected to the PostGIS database to expose spatial layers via OGC-compliant services—specifically WMS (Web Map Service) and WFS (Web Feature Service)**.** Data layers are styled, filtered, and organized into thematic groups (e.g., by country, gender, age group) to enhance usability. Each service endpoint is tested for stability, responsiveness, and cross-platform compatibility. In parallel, metadata are published to a catalog interface to support data discovery and citation. Comprehensive documentation—including API instructions and usage guidelines—is compiled to support external users such as policymakers, educators, and researchers. This phase ensures that the spatial data infrastructure is not only functional but also accessible and reusable.

Spatial Data Infrastructure

A **Spatial Data Infrastructure (SDI)** refers to the integrated framework of technologies, data, standards, policies, and human resources that enables the **effective collection, management, sharing, and use of geospatial data**. In this project, the SDI forms the backbone of cross-border labour market analysis by integrating cleaned Eurostat datasets into a **PostGIS spatial database,** enriched with standardized geometries and ISO 19139-compliant metadata. The infrastructure supports open **WMS/WFS services** via **GeoServer**, ensuring that employment indicators can be accessed, visualized, and reused across platforms. By following international geospatial standards and leveraging open-source technologies, the SDI enables interoperability, reproducibility, and long-term usability for stakeholders including policymakers, educators, and researchers.

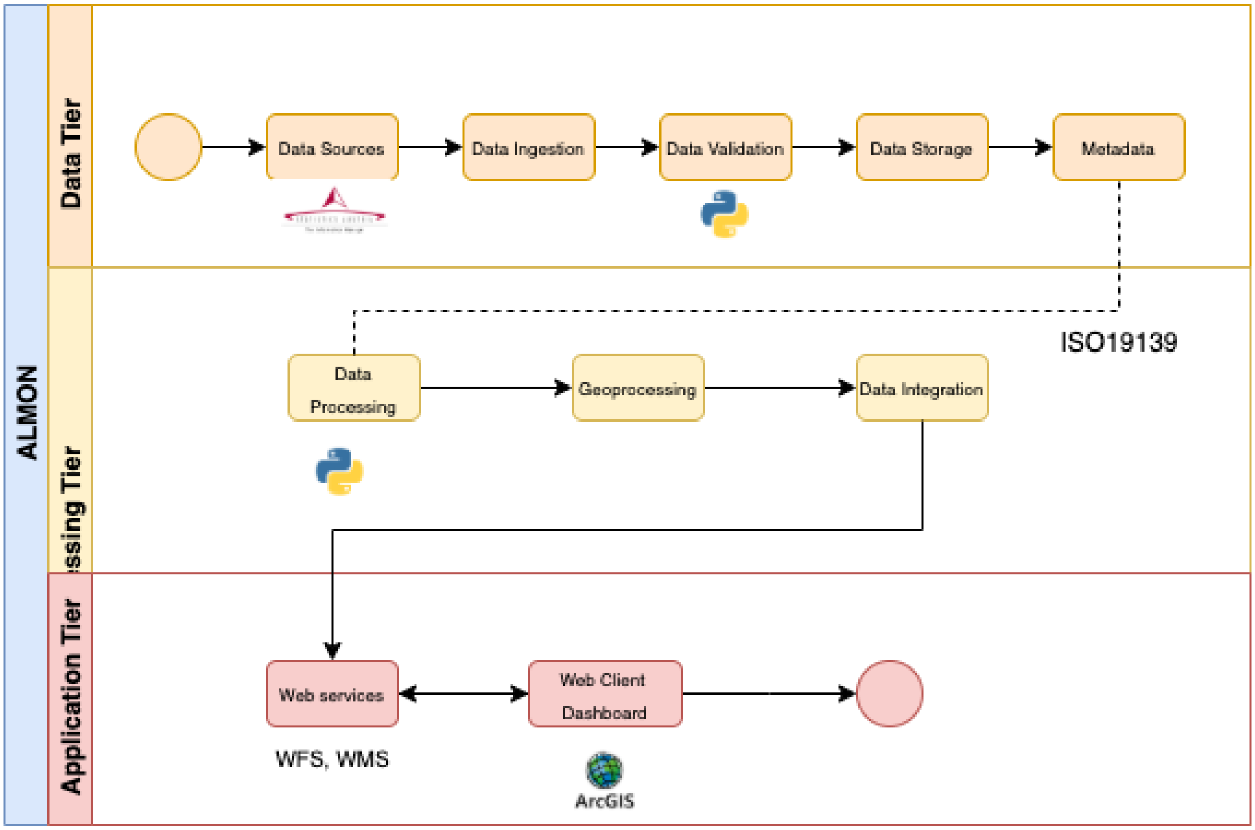
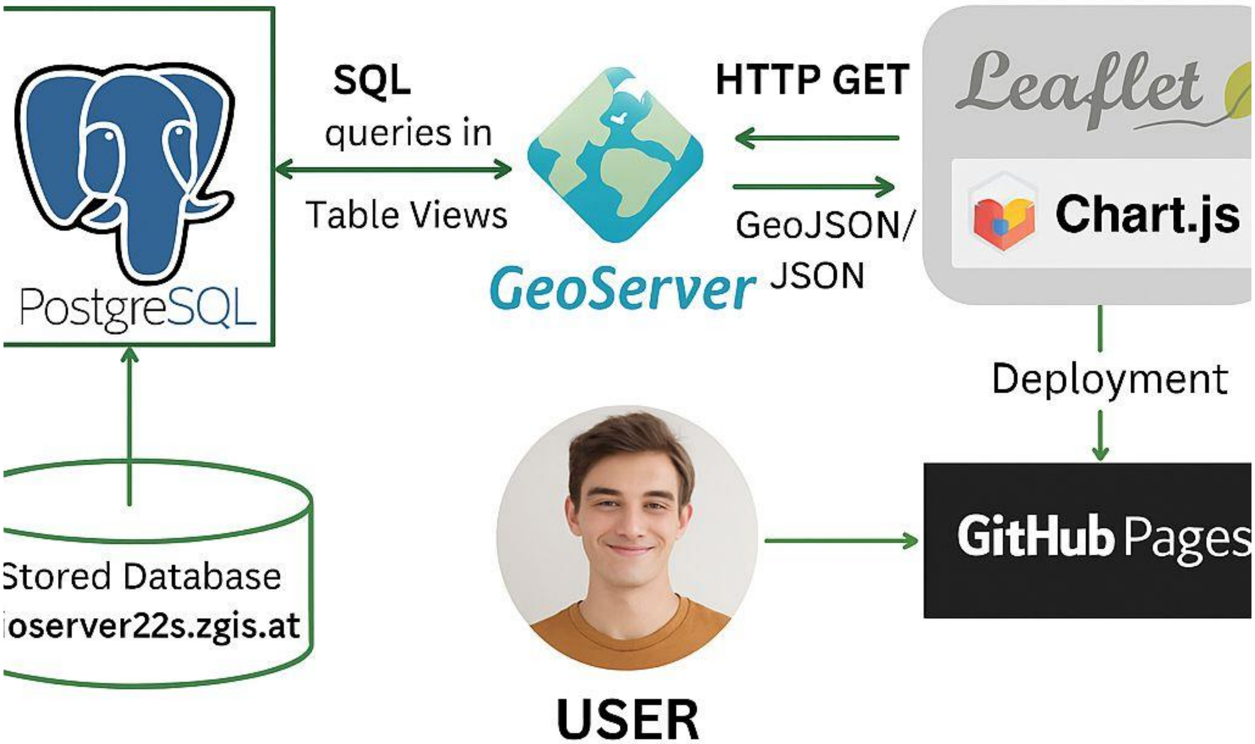


Figure 2: Spatial Data Infrastructure (SDI) overview



3 Milestones and deliverables

For each work package (WP), different milestones (Table 2) were set to keep track of the most important tasks that need to be completed in each work package. Similar to the milestones, deliverables were also defined for each work package (Table 3) to consolidate the expected outcomes for each WP.

Table 2: Milestones of project ALMON

	Name	Date Completion
M1	Project Planning and Workflow Setup	10.03.2025
M2	Data Collection and Pre-processing	24.03.2025
M3	Data Aggregation and Structuring	15.04.2025
M4	Database and Metadata Setup	10.05.2025
M5	Web Services Published	20.05.2025
M6	Dashboard Design Complete	10.06.2025
M7	Metadata Published	22.06.2025
M8	Project Presentation	25.06.2025
M9	Final Report Submitted	30.06.2025

Table 3: Deliverables and GitLab connections for project ALMON

	Name	Link
D1	**Project overview document**	https://git.sbg.ac.at/s1098705/almon/-/blob/main/Project%20management/Project_Overview_Template_en.docx?ref_type=heads
D2	**GitLab documentation**	https://git.sbg.ac.at/s1098705/almon
D3	**Final project report**	
D4	**Database with ready to publish data**	https://git.sbg.ac.at/s1098705/almon/-/tree/main/Metadata?ref_type=heads
D5	**Published dashboard**	https://euphoricking.github.io/employment-dashboard/

4 Results

The final dashboard is freely available at:

https://euphoricking.github.io/employment-dashboard/

The dashboard is adapted for desktop (Figure 3).

How to use the Dashboard

In the middle you can see an interactive map with a details instruction on how to use this map. On the left we have Austria, we have 'Year, Gender' selection. On the right we can select different countries.

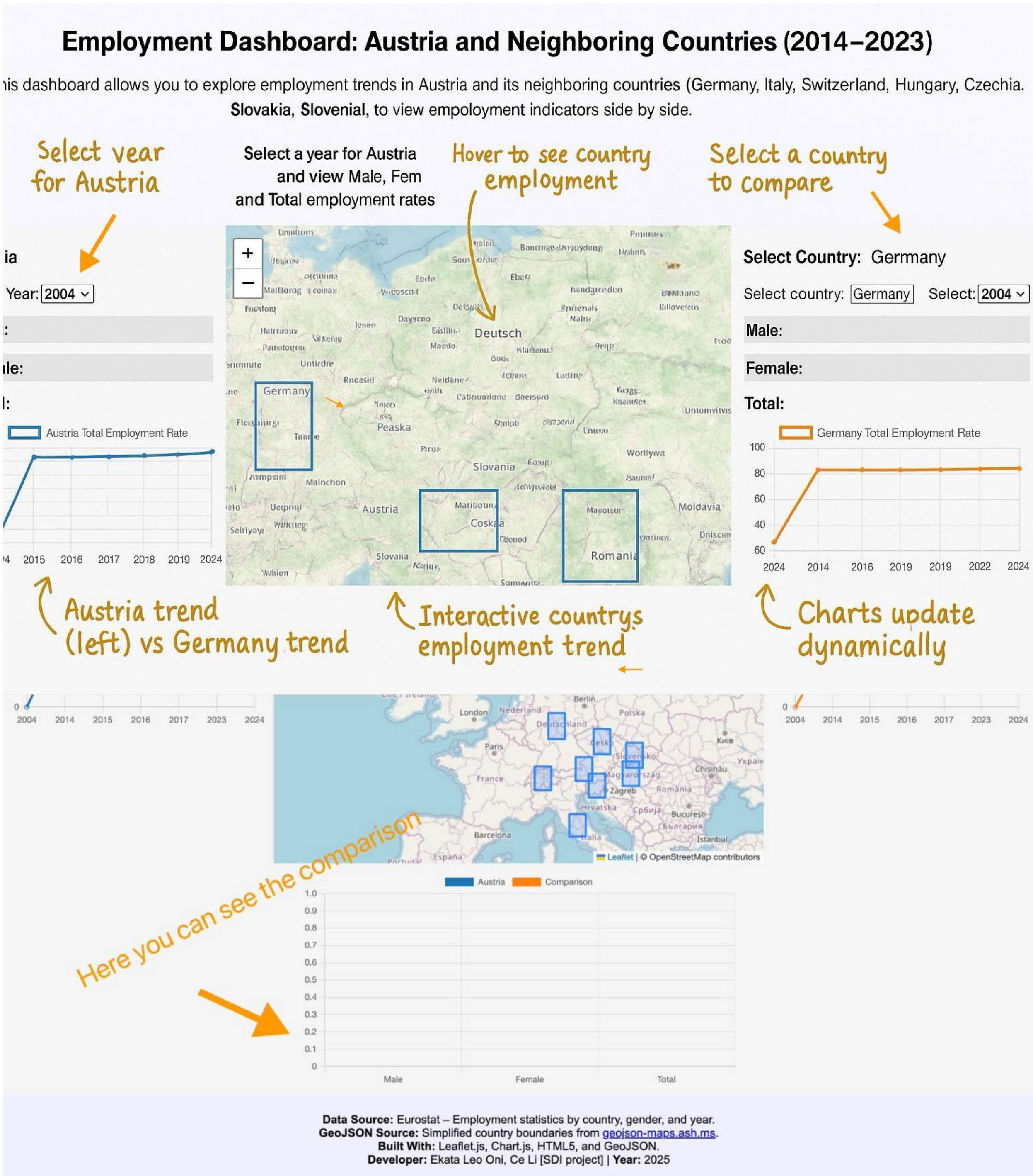


Figure 3: Desktop view of the final ALMON dashboard product

5 Discussion

This chapter reflects on the challenges, insights, and outcomes of the ALMON project, which aimed to develop a spatial data infrastructure (SDI) for analysing cross-border labour market trends. As a student-led initiative with a multidisciplinary background, the project presented both technical and organizational learning opportunities that extended beyond the initial scope.

The reason for us of using open source is because instead of relying on Esri's dashboard, open source have greater flexibility, transparency, and long-term sustainability.

Project management

A key lesson was the importance of **accurate time estimation and workflow coordination.** While the overall project was relatively small in scale, the complexity of coordinating between data acquisition, geospatial processing, and dissemination tools was initially underestimated. Setting up effective collaboration via GitLab and managing deliverables through defined milestones proved essential in maintaining progress. Particularly, the tasks related to **data cleaning and geocoding** required more effort than anticipated, underscoring the need for better time allocation and iterative validation in future projects.

Spatial Data Infrastructure

Spatial Data Infrastructure (SDI) refers to the framework that enables the collection, management, sharing, and utilization of geospatial data. It involves the coordination of data, technology, policies, and institutional frameworks to support spatial data usage in decision-making processes. SDI typically includes standards for data formats, metadata, and services, such as Web Mapping Services (WMS) or Web Feature Services (WFS), to facilitate interoperability across various platforms and organizations. The goal of SDI is to provide accessible and standardized geospatial data to support diverse applications in sectors like urban planning, environmental management, transportation, and disaster response. By ensuring that data is easily discoverable, shareable, and usable, SDI enhances informed decision-making, fosters collaboration, and promotes the effective use of spatial information for research, policy-making, and public services.